Sohaib Hasan

Personal Website | +92-3335904314 | Email | LinkedIn | GitHub | Chakwal, Pakistan

Summary

- Selected **Section Leader** for **Stanford University's Probability for AI** course, teaching international students.
- **9+ years** teaching advanced mathematics to undergraduate students.
- **Top 12% globally** in AI Mathematical Olympiad Progress Prize 3 (495th/4,138 teams, score 42/50).
- **Perfect score (10/10)** at Harvard CS50x Puzzle Day 2026 (166+ countries).
- Actively participating in international hackathons and building production-grade AI systems using LLMs, agentic architectures, and full-stack ML pipelines.
- Creator of PlotLab (**Instagram** and **Facebook**) — Manim-based mathematical animation channels (**100K+ Followers**), making advanced mathematics accessible through engaging visualizations.

Education

M.Phil in Mathematics **2018 – 2021**
Capital University of Science & Technology (CUST) *Islamabad, Pakistan*
- CGPA: 3.40/4.00
- Thesis: *Generalized Signcryption Based on Elliptic Curve Cryptography* (Link)

M.Sc in Mathematics **2015 – 2017**
COMSATS University Islamabad *Attock, Pakistan*
- Thesis: *Flow of Eyring-Powell Fluid in a Channel Using Adomian Decomposition Method* (Link)

B.Sc in Mathematics **2012 – 2014**
Horizon Degree & Commerce College *Chakwal, Pakistan*
- Core Coursework: Calculus, Linear Algebra, Mathematical Methods, Metric Spaces, Mechanics

Academic & Professional Experience

Section Leader (Volunteer), Probability for AI (PAI) | Course Website **Oct 2026 – Nov 2026**
Stanford University *Online (Global)*
- Selected from **2,700+ applicants** across **70 countries** as **Section Leader** for Stanford's *Probability for AI*.
- Lead a six-week section for **international students** in a course by Profs. Chris Piech and Mehran Sahami.
- Teach the probability behind modern AI through small-group, human-centred, hands-on learning.

Head of Department, Mathematics | Faculty Profile **June 2024 – Present**
Horizon Degree & Commerce College *Chakwal, Pakistan*
- Lead academic operations across multiple campuses: faculty assignments, course distribution, and curriculum development.
- Chair departmental seminars on emerging technologies in mathematics and AI, maintaining faculty alignment with current research developments.
- Serve as primary liaison between administration, faculty, and students to sustain instructional quality and resolve departmental challenges.

Lecturer in Mathematics **March 2017 – May 2024**
Horizon Degree & Commerce College *Chakwal, Pakistan*
- Over 9 years of teaching; taught undergraduate courses including Real Analysis, Measure Theory, Calculus, Functional Analysis, Linear Algebra, Discrete Mathematics, Number Theory, Statistics, and Machine Learning.
- Developed LaTeX-typeset course materials and Manim-animated visual lectures to strengthen conceptual understanding.
- Supervised undergraduate research projects and provided mentorship on independent mathematical inquiry.

International Competitions & Hackathons

AI Mathematical Olympiad (AIMO) — Progress Prize 3 | Kaggle **Nov 2025 – Apr 2026**

- **Top 12% globally** — Our 2-members team ranked **495/4,138 teams**, scoring **42/50 (84%)** on original IMO-standard problems from a $10M global prize fund competition.
- Engineered a weighted entropy pipeline on GPT-OSS-120B with Tool-Integrated Reasoning (TIR) and a parallel solving method of 8 simultaneous attempts per problem within a single inference pass.
- Optimised deployment on H100 GPUs (**Fields Model Initiative**); applied systematic hyperparameter tuning, mathematical guardrails, and few-shot prompting to guide model reasoning.

Harvard University CS50x Puzzle Day | Certificate **April 2026**

- **Perfect score (10/10 puzzles)** including the Metapuzzle — Harvard's annual international logic and combinatorial reasoning competition with participants from **166+ countries**.
- Demonstrated formal reasoning, combinatorial problem-solving, and systematic logical deduction under timed conditions.

RAISE YOUR HACK | *Official Hackathon of RAISE Summit 2025, Paris* | Certificate | GitHub **July 2025**

- Among **6,247 global participants** ($150,000 prize pool); built **Acumen AI** — a multi-agent conversational platform powered by MCP Agents, Groq, LLaMA-3.3-70B, and FAISS semantic search.
- Designed real-time chat interface with persistent memory, session management, and natural language querying for business intelligence applications.

Trae AI IDE: Zero Limits Hackathon | Certificate | GitHub **June 2025**

- Built an **AI-Powered Anti-Short-Video-Addiction Assistant** in a 48-hour global competition — XGBoost behavioral classification with real-time empathetic interventions.
- Architected a full-stack multi-agent pipeline: FastAPI AI inference layer (Python), Node.js frontend, Hugging Face and Novita AI model integration.

Selected Projects

Classroom AI Assistant | GitHub | Student App | Teacher Dashboard | *Python, Streamlit, Gemini API, RAG, SymPy*

- Build a Retrieval-Augmented Generation (**RAG**) AI system that grounds answers in curated course material using **Gemini embeddings**.
- Integrated **SymPy** for independent verification of AI-generated mathematical calculations.
- Developed dual **Streamlit** apps for student assistance and teacher analytics, with shared database integration and reliable API handling.

Socioeconomic Determinants of Housing Valuation | GitHub | *Python, Linear Regression*

- Empirical regression study quantifying income, structural, and demographic influences on property prices ($R^2 = 0.92$); emphasis on statistical interpretability and mathematical rigor.

Employee Attrition & Workforce Analytics | GitHub | *Power BI, Statistical Analysis*

- Analyzed 1,470-employee dataset (16.2% attrition rate); identified highest-risk cohorts and developed interactive Power BI dashboards to inform HR strategy.

Music Store Customer Behavior Analysis | GitHub | *PostgreSQL, Business Intelligence*

- Designed 11 structured SQL queries across a relational database to extract revenue patterns, genre preferences, and regional customer segmentation for strategic decision support.

Technical Skills

Programming & Typesetting: Python, SQL, LaTeX, Manim
Scientific Computing: NumPy, SymPy, Pandas, Matplotlib, Seaborn
Machine Learning: Scikit-learn, XGBoost, Hugging Face Transformers, FastAPI
LLMs & Agentic AI: Groq API, Gemini API, LLaMA (3.3-70B), Retrieval-Augmented Generation (RAG), MCP (Model Context Protocol), FAISS, Tool-Integrated Reasoning (TIR), Monte Carlo Tree Search (MCTS)
Data & Visualization: Excel, Tableau, Power BI, MySQL, PostgreSQL
Platforms: VS Code, Trae AI IDE, Jupyter Notebook, Google Colab, Streamlit, Git/GitHub

References

Dr. Rashid Ali **rashid.ali@cust.edu.pk** | **+92-332-506-5505**

- Professor & M.Phil Thesis Supervisor, Dept. of Mathematics, Capital University of Science & Technology, Islamabad

Dr. Ali Imran **ali.imran@cuiatk.edu.pk** | **+92-331-908-7342**

- Assoc. Professor (Tenured) & M.Sc Thesis Supervisor, Dept. of Mathematics, COMSATS University Islamabad, Attock

Dr. Shafiq-ur-Rehman **shafiq@cuiatk.edu.pk** | **+92-300-812-9602**

- Assoc. Professor (Tenured), Dept. of Mathematics, COMSATS University Islamabad, Attock